DATASCI 406 FA 26, Quiz 1.3

Name and Uniqname:

Question 1 (1 point)

To estimate the bias of $\hat{\theta}$ you simulate k datasets of size n and run `t.test(theta_hats - theta)`. You rerun with a much larger k, leaving n unchanged. What gets smaller? **Circle one.**

1. Var $\left(\hat{\theta}\right)$ — the estimator itself becomes more precise.
2. The `t.test` interval around the estimated bias — we pin the bias down better.
3. Bias$(\hat{\theta})$ — the estimator itself becomes less biased.
4. All three: a longer run improves every number the simulation reports.

Question 2 (1 point)

You simulate to check whether `t.test(x)$conf.int` really covers 95% of the time for skewed data. Which ingredient does the simulation need that a real data analysis can never supply? **Circle one.**

1. The true value of θ, so we can check whether the interval caught it.
2. A sample size n matching the dataset we actually have.
3. Enough replications k that the reported coverage is precise.
4. A confidence level $1 - \alpha$ fixed before looking at the data.

Question 3 (3 points)

$X_1, \ldots, X_{10}$ are i.i.d. $N(\mu, \sigma^2)$ with $\sigma^2 = 1$. Two estimators of σ^2, run on the same $k = 100{,}000$ simulated datasets:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \qquad \hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

	bias	variance
S^2	0.00	0.22
MLE	-0.10	0.18

Circle the estimator you would use and justify it in two or three sentences. You must say **both** (i) which single quantity ranks the two, and why neither column settles it alone, and (ii) what that quantity is for each estimator.

Circle: **S^2** or **MLE**